

Informal Question (Section 2)

Unrefereed candidate disproof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 1812.09215__00 — generated 2026-09-17

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result would_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

Summary and provenance

Catalog id	1812.09215__00
Catalog entry	Informal Question (Section 2)
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Lipschitz bijections between boolean functions
Model verdict	disproved (confidence: high)
Model's one-line claim	No: every such bijection satisfies $n \leq 1 + (k-1) + \dots + (k-1)^{(D-1)}$, where D is its inverse Lipschitz constant, and linear 2-Lipschitz tree maps attain the resulting bound.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/1812.09215__00/output.md
Independent review	None
Model's caveats	Uses full-cube bijections and global k-junta locality; literature priority has not been independently checked.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

A sharp negative answer

The gap closes on the lower-bound side. In fact, the forward Lipschitz assumption is unnecessary for the lower bound, and a rooted-tree construction gives the exact optimum.

The proof below does not rely on the previous attempt's localized-dependency estimate. It retains the elementary inverse-edge setup, verified below, but replaces dependency-graph packing by a bounded-branching count of possible target outputs.

1. Statement

Let $Q_n = \{0, 1\}^n$, equipped with Hamming distance. A Dictator-to-XOR bijection is a bijection

$$\phi : Q_n \longrightarrow Q_n$$

such that

$$\bigoplus_{a=1}^n \phi_a(x) = x_1 \quad \text{for every } x \in Q_n. \quad (1)$$

Assume that, for each output coordinate a , there is a set $P_a \subseteq [n]$, with $|P_a| \leq k$, such that ϕ_a depends only on the input coordinates in P_a . Write

$$D = \text{Lip}(\phi^{-1}).$$

On a Hamming cube, the Lipschitz constant equals the maximum stretch of an edge, so D is an integer.

For $k \geq 2$ and $d \geq 1$, put

$$N_k(d) = \sum_{r=0}^{d-1} (k-1)^r.$$

Theorem Every bijection satisfying (1) and the locality condition obeys

$$\boxed{n \leq N_k(D)}. \quad (2)$$

Conversely, whenever $n \leq N_k(d)$, there is a **linear, 2-Lipschitz** Dictator-to-XOR bijection whose output coordinates depend on at most k inputs and whose inverse is d -Lipschitz.

Consequently, the minimum possible inverse Lipschitz constant is exactly

$$\boxed{D_{\min}(n, k) = \begin{cases} n, & k = 2, \\ \left\lceil \frac{\log((k-2)n+1)}{\log(k-1)} \right\rceil, & k \geq 3. \end{cases}} \quad (3)$$

This remains the exact optimum if any fixed forward Lipschitz bound $C \geq 2$ is imposed.

In particular,

$$\boxed{D \geq 1 + \frac{\log n}{\log k}}. \quad (4)$$

Thus the inverse cannot be $o(\log n / \log k)$ -Lipschitz.

2. The counting argument

We prove a slightly stronger pointwise statement.

Fix an input $x \in Q_n$, and write $y = \phi(x)$. For each output coordinate a , define

$$x^{(a)} = \phi^{-1}(y \oplus e_a), \quad R_a = \{i : x_i \neq x_i^{(a)}\}.$$

The sets R_a are nonempty and pairwise distinct.

Equation (1) gives

$$x_1^{(a)} = \bigoplus_b (y \oplus e_a)_b = \left(\bigoplus_b y_b \right) \oplus 1 = x_1 \oplus 1.$$

Hence

$$1 \in R_a \quad \text{for every } a. \quad (5)$$

For a set $T \subseteq [n]$, let

$$x^T = x \oplus \bigoplus_{i \in T} e_i.$$

Thus $x^{R_a} = x^{(a)}$.

Fix an integer $d \geq 1$. For nonempty $T \subseteq [n]$, with $|T| \leq d$, define the family of target labels

$$A_d(T) = \{a : T \subseteq R_a, |R_a| \leq d\}.$$

Claim For every such T ,

$$|A_d(T)| \leq N_k(d - |T| + 1). \quad (6)$$

Proof We use induction on $d - |T|$.

If $|T| = d$, every $a \in A_d(T)$ must satisfy $R_a = T$. Since the R_a are distinct,

$$|A_d(T)| \leq 1 = N_k(1).$$

Now suppose $|T| < d$. Because $T \neq \emptyset$ and ϕ is injective,

$$\phi(x^T) \neq \phi(x).$$

Choose an output coordinate b on which these two images differ:

$$\phi_b(x^T) \neq \phi_b(x). \quad (7)$$

Since ϕ_b depends only on P_b , equation (7) implies

$$P_b \cap T \neq \emptyset, \quad \text{and therefore} \quad |P_b \setminus T| \leq k - 1. \quad (8)$$

Consider any target label $a \in A_d(T)$ with $a \neq b$. By the definition of R_a ,

$$\phi(x^{R_a}) = \phi(x) \oplus e_a.$$

Since $a \neq b$, it follows that

$$\phi_b(x^{R_a}) = \phi_b(x) \neq \phi_b(x^T). \quad (9)$$

The inputs x^{R_a} and x^T differ precisely on $R_a \setminus T$, because $T \subseteq R_a$. Locality and (9) therefore imply

$$P_b \cap (R_a \setminus T) \neq \emptyset.$$

Thus a belongs to $A_d(T \cup \{i\})$ for at least one $i \in P_b \setminus T$.

We have proved the covering relation

$$A_d(T) \subseteq \{b\} \cup \bigcup_{i \in P_b \setminus T} A_d(T \cup \{i\}). \quad (10)$$

The exceptional set $\{b\}$ has size one; it does not matter whether b actually belongs to $A_d(T)$.

Applying the induction hypothesis to every child set $T \cup \{i\}$, and using (8), gives

$$\begin{aligned} |A_d(T)| &\leq 1 + \sum_{i \in P_b \setminus T} |A_d(T \cup \{i\})| \\ &\leq 1 + (k - 1)N_k(d - |T|) \\ &= N_k(d - |T| + 1). \end{aligned}$$

This completes the induction. $\square$

Applying the claim By (5), every inverse edge at y changes input coordinate 1. Therefore

$$A_d(\{1\}) = \{a : |R_a| \leq d\}.$$

Taking $T = \{1\}$ in (6), we obtain the pointwise bound

$$\# \{a : d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq d\} \leq N_k(d). \quad (11)$$

For $d = D$, all n inverse edges have stretch at most D . Equation (11) yields

$$n \leq N_k(D),$$

proving (2).

The essential distinction from the earlier packing argument is this: at a partial input change T , choose **one** currently changed output b . Its own target label accounts for at most one possibility. Every other target must add an input from the set $P_b \setminus T$, which has size at most $k - 1$. This gives a branching factor $k - 1$, without any dependence on the radius d .

3. Sharpness: the rooted-tree construction

Let T be a rooted tree with n vertices, root 1, maximum number of children at most $k - 1$, and height at most $d - 1$.

Such a tree exists whenever

$$n \leq 1 + (k - 1) + \dots + (k - 1)^{d-1} :$$

take the first n vertices in breadth-first order in the complete rooted $(k - 1)$ -ary tree of height $d - 1$.

Index both input and output coordinates by the vertices of T . Define

$$\phi_v(x) = x_v \oplus \bigoplus_{w: w \text{ is a child of } v} x_w. \quad (12)$$

We verify all the required properties.

Locality Each ϕ_v depends on its own input and on its children's inputs, hence on at most

$$1 + (k - 1) = k$$

input coordinates.

Dictator-to-XOR condition In the XOR of all outputs, the root input appears once. Every nonroot input appears twice: in its own output and in its parent's output. Consequently,

$$\bigoplus_v \phi_v(x) = x_1.$$

Bijectivity The inverse is

$$x_v = \bigoplus_{w \in \text{subtree}(v)} y_w, \quad y = \phi(x). \quad (13)$$

Indeed, summing (12) over the subtree rooted at v cancels every input in that subtree except x_v . Thus every input is recovered uniquely.

Forward Lipschitz constant Flipping the root input changes only the root output. Flipping a nonroot input x_v changes exactly the outputs indexed by v and its parent. Hence

$$\text{Lip}(\phi) \leq 2.$$

Inverse Lipschitz constant By (13), flipping output y_w changes exactly those inputs indexed by ancestors of w , including w . Its inverse stretch is therefore

$$\text{depth}(w) + 1 \leq d.$$

Thus

$$\text{Lip}(\phi^{-1}) \leq d.$$

This proves the converse assertion. Choosing the smallest d with $n \leq N_k(d)$ proves the exact optimum (3).

For completeness, when $n \geq 2$, a forward Lipschitz bound below 2 is impossible: flipping an input coordinate other than 1 preserves output parity, so injectivity forces the output Hamming distance to be a positive even integer. Therefore the construction also uses the smallest possible forward Lipschitz constant.

4. Consequences for the question

For $k = 2$, equation (2) is simply

$$n \leq D.$$

For $k \geq 3$,

$$n \leq \frac{(k-1)^D - 1}{k-2},$$

which gives the second line of (3).

A convenient uniform consequence follows from

$$\sum_{r=0}^{D-1} (k-1)^r \leq \sum_{r=0}^{D-1} \binom{D-1}{r} (k-1)^r = k^{D-1}.$$

Thus

$$n \leq k^{D-1}, \quad \text{so} \quad D \geq 1 + \frac{\log n}{\log k}.$$

This excludes

$$D = o\left(\frac{\log n}{\log k}\right)$$

along every sequence with $n, k \rightarrow \infty$.

The degenerate locality case $k = 1$ permits no such bijection for $n \geq 2$: in (10), $P_b \setminus T$ is empty, so taking $T = \{1\}$ gives $n \leq 1$.

Conclusion. Under the standard full-cube and global-junta definitions in the question, nonlinear maps offer no improvement in the optimal maximum inverse stretch: linear 2-Lipschitz rooted-tree maps are already exactly optimal. No additional conjecture or unproved mathematical step is used. The remaining caveat is literature priority, which has not been independently checked here.